Equipment state [REST]

Actuator state + occupancy [REST]

Room Climate Control [System] (one Compose project)

…/sensor [MQTT]

Per-room device containers (5 per controlled room)

…/ac [MQTT]

Occupancy publisher
[Container: Python]
con-occ.py
Copies head-count into BuildSim

Temperature + CO_2 sensors
[2 Containers: Python]
con-sensor.py
Write reading to BuildSim; forward

HVAC + ventilation actuators
[2 Containers: Python]
con-hvac.py, con-vent.py
Apply command; acknowledge

Physical simulator
[Container: Python]
con-sim.py (one for all rooms)
Heat + CO_2 balance per room

Dashboard
[Container: Grafana 11]
Room time series

Ack …/actuator-status [MQTT]

Reading …/dec [MQTT]

Physical value [MQTT]

PromQL [HTTP]

Message broker
[Container: Mosquitto 2]
Topic per room / signal / hop

Prometheus
[Container: Prometheus 2.54]
5 s scrape; tmpfs storage

Readings + acks [MQTT] Commands …/ac [MQTT]

Scrape /metrics [HTTP]

Write occupancy sensor [REST]

Write sensor value [REST]

Decision maker
[Container: Python]
con-dec.py (one for all rooms)
2 PID loops per room + safety

Metrics exporter
[Container: Python]
metrics-exporter.py
BuildSim state as Prometheus text

Wait for 5 sim-min step [REST]

Set actuator state [REST]

Clock for setpoint [REST]

Read head-counts [REST]

BuildSim
[Software System]
Sensor values, actuator state, 3D viewer

Occupancy simulator
[Software System]
Simulated clock + head-count per room